

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1695

Country Club 3 100 kV
February 22, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1695. The Interconnection Customer proposes a 288.9 MW solar photovoltaic facility interconnecting at the Country Club 3 100 kV substation in the Indiana area. The request was received on March 15, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 615 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 15 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$54,550,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Montpelier Electric Generating Station Substation 100/1kV TX	Monfort-Wynedale 230 KV	179.4	257	66.6
Bedford Rural-Hartford City 100 KV Line	Montpelier Electric Generating Sta 230/1kV Auto	160.5	136	42.1
Hartford City-Intalco 100 KV Line	Signetics-Montpelier Electric Generating Sta 100 KV	157.5	104	25.9
Intalco-Montpelier Electric Generating Station Substation 100 KV line	Montpelier Electric Generating Sta-Signetics 100 KV	149.4	234	59.7
Montpelier Electric Generating Sta 230/1kV Auto	Waynedale-Monfort Substation 230 KV	147.5	313	66.6
Hartford City-Signetics 100 KV	Signetics-Montpelier Electric Generating Station 100 KV	145.3	309	72.3
Signetics-Montpelier Electric Generating Sta 100 KV	Hartford City-Signetics 100 KV	140.4	372	84.8
Hfman-Intalco 100 KV Line	Signetics-Country Club 3 100 KV line	124.9	184	45.4
Hofman-Hartford Ciy 100 KV L	Country Club 3-Signetics 100 KV L	124.3	211	54.6
Randville-Hofman 100 KV	Signetics-Country Club 3 Substation 100 KV L	122.9	415	100.0
Randville-Bluffton 100 KV	Signetics-Country Club 3 100 KV	121.4	436	100.0
Bluffton Substation-Azle 100 KV	Signetics-Country Club 3 Substation 100 KV	120.2	453	100.0

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Signetics Substation-Country Club 3 100 KV	Azle-Country Club 3 100 KV	119.0	474	100.0
Azle Substation-Country Club 3 100 KV	Signetics-Country Club 3 Substation 100 KV	119.0	474	100.0
Bedford Rural-China 100 KV	Montpelier Electric Generating Station 230-1 kV Transformer Bank	111.8	233	55.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Montpelier Electric Generating Station Substation 100/1kV TX	Replace transformer with higher-rated unit	\$2,891,000	43% (\$1,231,000)
Bedford Rural-Hartford City 100 KV Line	Rebuild line with higher-capacity conductor	\$4,141,000	48% (\$1,979,000)
Hartford City-Intalco 100 KV	Rebuild line with higher-capacity conductor	\$4,276,000	52% (\$2,218,000)
Montpelier Electric Generating Sta-Intalco 100 KV	Rebuild line with higher-capacity conductor	\$10,238,000	51% (\$5,262,000)
Montpelier Electric Generating Station 230/1 kV Transformer Bank	Replace transformer with higher-rated unit	\$9,080,000	43% (\$3,866,000)
Hartford City-Signetics 100 KV	Rebuild line with higher-capacity conductor	\$12,087,000	52% (\$6,222,000)
Signetics-Montpelier Electric Generating Sta 100 KV L	Rebuild line with higher-capacity conductor	\$2,053,000	52% (\$1,060,000)
Hfman-Intalco 100 KV Line	Rebuild line with higher-capacity conductor	\$6,297,000	62% (\$3,939,000)
Hartford City-Hofman 100 KV	Rebuild line with higher-capacity conductor	\$3,093,000	62% (\$1,934,000)
Randville-Hofman 100 KV	Rebuild line with higher-capacity conductor	\$18,515,000	62% (\$11,580,000)
Randville-Bluffton 100 KV	Rebuild line with higher-capacity conductor	\$6,020,000	62% (\$3,765,000)
Bluffton-Azle 100 KV line	Rebuild line with higher-capacity conductor	\$1,920,000	62% (\$1,201,000)
Country Club 3-Signetics 100 KV	Rebuild line with higher-capacity conductor	\$4,859,000	62% (\$3,039,000)
Azle-Country Club 3 100 KV L	Rebuild line with higher-capacity conductor	\$6,933,000	62% (\$4,336,000)
Bedford Rural-China 100 KV	Rebuild line with higher-capacity conductor	\$6,389,000	46% (\$2,918,000)
	Total allocated to project		\$54,550,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.